

UNIT 3: LINEAR TRANSFORMATIONS AND INNER PRODUCT SPACES

Chapter 42: Matrix and linear transformations

Definition: Let V and W be vector spaces. A function $T: V \rightarrow W$ is called a **linear transformation** of V into W if

(a) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for every $\mathbf{u}, \mathbf{v}$ in V .

(b) $T(c\mathbf{u}) = cT(\mathbf{u})$ for every $\mathbf{u}$ in V and c in $\mathbb{R}$.

V is called the **domain space** and W is called the **codomain space** of T .

For $T: V \rightarrow W$, $\mathbf{w}$ is the **image** of a vector $\mathbf{v}$ if $T(\mathbf{v}) = \mathbf{w}$.

A linear transformation $T: V \rightarrow V$ is also called a **linear operator** on V .

Example: Consider $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_3 \\ x_2 \end{bmatrix}$. Show this is a linear transformation.

$$\begin{aligned} \text{(a)} \quad \forall \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, \quad T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\right) &= \begin{bmatrix} x_1 + x_3 + y_1 + y_3 \\ x_2 + y_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 + x_3 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 + y_3 \\ y_2 \end{bmatrix} \\ &= T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) + T\left(\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\right) \\ \text{(b)} \quad T\left(c \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) &= \begin{bmatrix} cx_1 + cx_3 \\ cx_2 \end{bmatrix} = c \cdot T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) \end{aligned}$$

Definition: A function $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by multiplication by a matrix A so that $T(\mathbf{x}) = A\mathbf{x}$ is called a **matrix transformation**. In this situation, we write T as T_A .

Example: Is $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_3 \\ x_2 \end{bmatrix}$ a matrix transformation?

Yes, $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

Theorem: Every matrix transformation is a linear transformation.

Proof:

$$T_A(x) = A \cdot x.$$

$$(a). \forall x, y, \quad T_A(x+y) = A \cdot (x+y) = A \cdot x + A \cdot y = T_A(x) + T_A(y)$$

$$(b) \quad T_A(c \cdot x) = A(c \cdot x) = cA \cdot x = cT_A(x)$$

Remark: If $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a matrix transformation (i.e. $T(x) = Ax$, then A is a m by n matrix.

Example: For any vector spaces V and W , the function $Z : V \rightarrow W$ by $Z(v) = \mathbf{0}_W$ is a linear transformation.

$$\forall x, y \in V, \quad Z(x+y) = \vec{0}_W = Z(x) + Z(y)$$

$$Z(c \cdot x) = \vec{0}_W = c \cdot Z(x)$$

Example: For any vector space V , the function $I_V : V \rightarrow V$ by $I_V(v) = v$ is a linear transformation (called the **identity operator**).

Example: Let $L : P_2 \rightarrow P_2$ with $L(p(x)) = xp'(x)$. Is L a linear transformation?

$$\begin{aligned} \text{Yes, } (a) \quad \forall p_1, p_2 \in P_2, \quad L(p_1 + p_2) &= x \cdot (p_1 + p_2)' \\ &= x \cdot p_1' + x \cdot p_2' \\ &= L(p_1) + L(p_2) \end{aligned}$$

$$(b) \quad L(c \cdot p_1) = x(c \cdot p_1)' = c \cdot xp_1' = c \cdot L(p_1)$$

Example:

Is $D : M_{22} \rightarrow \mathbb{R}$ by $D(A) = \det(A)$ is a linear transformation?

No. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

$$D(A+B) = 0 \neq 2 = D(A) + D(B)$$

Is $S : M_{22} \rightarrow \mathbb{R}$ by $S(A) = \text{tr}(A)$ is a linear transformation?

Yes. (a) $\forall A, B \in M_{2 \times 2}$. $S(A+B) = a_{11} + b_{11} + a_{22} + b_{22} = S(A) + S(B)$

$$(b) S(c \cdot A) = ca_{11} + ca_{22} = c \cdot S(A)$$

Theorem: Let $T : V \rightarrow W$ be a linear transformation. Then

(a) $T(\mathbf{0}_V) = \mathbf{0}_W$.

(b) $T(-\mathbf{v}) = -T(\mathbf{v})$ for $\mathbf{v}$ in V .

(c) $T(r_1 \mathbf{v}_1 + r_2 \mathbf{v}_2 + \cdots + r_k \mathbf{v}_k) = r_1 T(\mathbf{v}_1) + r_2 T(\mathbf{v}_2) + \cdots + r_k T(\mathbf{v}_k)$ for all $\mathbf{v}_i$ in V and scalar r_i .

Proof: (a) and (b) (try (c) on your own)

(a). Let $c \neq 1$. then $\frac{T(c \cdot \vec{0}_V)}{T(\vec{0}_V)} = c \cdot \frac{T(\vec{0}_V)}{T(\vec{0}_V)}$

$\Rightarrow (c-1) \cdot T(\vec{0}_V) = \vec{0}_W$, since $c-1 \neq 0 \Rightarrow T(\vec{0}_V) = \vec{0}_W$.

(b) $T(\vec{v}) + T(-\vec{v}) = T(\vec{0}_V) = \vec{0}_W \Rightarrow T(-\vec{v}) = -T(\vec{v})$

Theorem: Let $T: V \rightarrow W$ be a linear transformation. Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis for V . Then T is uniquely determined by $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$.

Proof:

$$\forall \mathbf{v} \in V, \quad \mathbf{v} = a_1 \mathbf{v}_1 + \dots + a_n \mathbf{v}_n.$$

$$\begin{aligned} \Rightarrow T(\mathbf{v}) &= T(a_1 \mathbf{v}_1 + \dots + a_n \mathbf{v}_n) \\ &= a_1 T \mathbf{v}_1 + \dots + a_n T \mathbf{v}_n \end{aligned}$$

Example: Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation and let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be a basis for $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ -5 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}.$$

Suppose that

$$T(\mathbf{v}_1) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad T(\mathbf{v}_2) = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \quad T(\mathbf{v}_3) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Find $T\left(\begin{bmatrix} 16 \\ 9 \\ 10 \end{bmatrix}\right)$.

$$\begin{aligned} T\left(\begin{bmatrix} 16 \\ 9 \\ 10 \end{bmatrix}\right) &= T(3\mathbf{v}_1 - 2\mathbf{v}_2 + 8\mathbf{v}_3) = 3T\mathbf{v}_1 - 2T\mathbf{v}_2 + 8T\mathbf{v}_3 \\ &= 3\begin{bmatrix} 1 \\ 2 \end{bmatrix} - 2\begin{bmatrix} 3 \\ 3 \end{bmatrix} + 8\begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -3 \\ -8 \end{bmatrix} \end{aligned}$$

Corollary: Let V and W be vector spaces and let $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis of V . Given any vectors $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n$ in W (they need not be distinct), there exists a unique linear transformation $T : V \rightarrow W$ satisfying $T(\mathbf{b}_i) = \mathbf{w}_i$ for each $i = 1, 2, \dots, n$. In fact, the action of T is as follows: Given $\mathbf{v} = r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \dots + r_n\mathbf{b}_n$ in V , r_i in $\mathbb{R}$, then

$$T(\mathbf{v}) = T(r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \dots + r_n\mathbf{b}_n) = r_1T(\mathbf{b}_1) + \dots + r_nT(\mathbf{b}_n)$$

Corollary: If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation, then there exists a unique $m \times n$ matrix A so that $T = T_A$, where $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is defined by $T_A(\mathbf{x}) = A\mathbf{x}$.

In this case, $T = T_A$ is the matrix transformation induced by the unique matrix A , given in terms of its columns by

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \dots \quad T(\mathbf{e}_n)],$$

where $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ are the standard basis vectors of $\mathbb{R}^n$.

A is called the **standard matrix of T** .

Example: Suppose I want to build a linear transformation T in $\mathbb{R}^2$ which is (counter-clockwise) rotation by 90° .

$$T\mathbf{e}_1 = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$T\mathbf{e}_2 = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$

$$\Rightarrow T_A(\mathbf{x}) = A \cdot \mathbf{x} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \cdot \mathbf{x}.$$

Chapter 43: Kernel and Image

Definition: Let $T: V \rightarrow W$ be a linear transformation.

The **kernel** of T , denoted by $\ker T$, is the subset of V consisting of all elements $\mathbf{v}$ of V such that $T(\mathbf{v}) = \mathbf{0}_W$.

The **image** of T denoted by $\operatorname{im} T$, consists of all vectors in W that are images under T of vectors in V . Thus $\mathbf{w}$ is in $\operatorname{im} T$ if there is a $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{w}$.

Remark: Some books refer to image as “range.”

In set notation:

$$\ker T = \{ \mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}_W \}$$

$$\operatorname{im} T = \{ \mathbf{w} \in W \mid \exists \mathbf{v} \in V, T(\mathbf{v}) = \mathbf{w} \}$$

Fact: $\ker T$ is never an empty set, because...

$$\mathbf{0}_V \in \ker T$$

$\operatorname{im} T$ is never an empty set, because...

$$\mathbf{0}_W \in \operatorname{im} T$$

Theorem: Let $T: V \rightarrow W$ be a linear transformation

1. $\ker T$ is a subspace of V .

2. $\operatorname{im} T$ is a subspace of W .

Proof: (Pick one)

proof of 2: (1) $\forall \mathbf{w}_1, \mathbf{w}_2 \in \operatorname{im} T, \exists \mathbf{v}_1, \mathbf{v}_2 \in V,$

$$\text{s.t. } T\mathbf{v}_1 = \mathbf{w}_1, T\mathbf{v}_2 = \mathbf{w}_2.$$

$$\Rightarrow T(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{w}_1 + \mathbf{w}_2 \Rightarrow \mathbf{w}_1 + \mathbf{w}_2 \in \operatorname{im} T.$$

$$(2). \text{ For } \mathbf{w}_1 \in \operatorname{im} T, T(c\mathbf{v}_1) = cT(\mathbf{v}_1)$$

$$= c\mathbf{w}_1$$

$$\Rightarrow c\mathbf{w}_1 \in \operatorname{im} T$$

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so $\operatorname{im} T$ is a subspace.

Definition: Given a linear transformation $T : V \rightarrow W$:

$\dim(\ker T)$ is called the **nullity** of T and is denoted $\text{nullity}(T)$.

$\dim(\text{im } T)$ is called the **rank** of T and is denoted $\text{rank}(T)$.

Example: Let $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the linear transformation induced by the $m \times n$ matrix A , i.e.,

$$T_A(\mathbf{x}) = \underline{A \cdot \mathbf{x}}$$

Then $\ker T_A = \underline{\text{null}(A)}$ and $\text{im } T_A = \underline{\text{im } A}$.

Write $A = [\mathbf{c}_1 \ \mathbf{c}_2 \ \cdots \ \mathbf{c}_n]$. Write out $\text{im } T_A$ in terms of the $\mathbf{c}_i$.

$$\text{im } T_A = \text{span} \{ \mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n \}.$$

This means: $\text{im } T_A = \text{col } A$, so $\text{rank}(T_A) = \underline{\text{rank}(A)}$.

Example: Let $L : P_2 \rightarrow \mathbb{R}$ be the function defined by

$$L(ax^2 + bx + c) = \int_0^1 (ax^2 + bx + c) dx.$$

(a) Why is L a linear transformation?

Yes. $p_1(x) = a_1x^2 + b_1x + c_1$

$$p_2(x) = a_2x^2 + b_2x + c_2$$

$$\begin{aligned} \text{(a)} \quad L(p_1 + p_2) &= \int_0^1 p_1 + p_2 dx = \int_0^1 p_1 dx + \int_0^1 p_2 dx \\ &= L(p_1) + L(p_2) \end{aligned}$$

$$\text{(b)} \quad L(c \cdot p_1) = \int_0^1 c \cdot p_1 dx = c \cdot \int_0^1 p_1 dx = c \cdot L(p_1)$$

(b) Find $\ker L$.

$$\begin{aligned} \forall p \in \ker L, & \Rightarrow L(p) = \int_0^1 p(x) dx = 0 \\ & = \int_0^1 (ax^2 + bx + c) dx = 0 \\ & \Rightarrow \frac{1}{3}a + \frac{1}{2}b + c = 0 \Rightarrow a = -\frac{3}{2}b - 3c \\ & \Rightarrow \ker L = \text{span} \left\{ -\frac{3}{2}x^2 + x, -3x^2 + 1 \right\} \end{aligned}$$

(c) Find $\dim \ker L = \text{nullity}(L)$.

$$\dim \ker L = 2.$$

Theorem: Let A be an $m \times n$ matrix, and let $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the linear transformation induced by A , that is $T_A(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$.

What if $\text{rank } A = r$? What can we say about the relationship between $\dim \text{Im } T_A$ and $\dim \ker T_A$?

$$\dim(\text{Im } T_A) = \text{rank}(T_A) = \text{rank}(A) = r$$

$$\dim(\ker T_A) = \dim(\text{null}(A)) = \text{nullity}(A) = n - r.$$

$$\Rightarrow \dim(\text{Im } T_A) + \dim(\ker T_A) = n.$$

It turns out that this result holds in general for all linear transformations of finite-dimensional vector spaces:

Theorem: (Dimension Theorem)

If $T : V \rightarrow W$ is a linear transformation between finite-dimensional vector spaces, then

$$\dim \ker T + \dim \text{Im } T = \dim(V).$$

In other words,

Example: Apply this theorem to $Z: V \rightarrow W$ by $Z(\mathbf{v}) = \mathbf{0}_W$.

$$\dim(\ker Z) = \dim(V) = n.$$

$$\dim(\operatorname{Im} Z) = \dim \langle \mathbf{0}_W \rangle = 0$$

$$n + 0 = n.$$

Example: Go back to $L: P_2 \rightarrow \mathbb{R}$ by $L(ax^2 + bx + c) = \int_0^1 (ax^2 + bx + c) dx$ and apply the Dimension Theorem.

Example: Consider $T: M_{2 \times 2} \rightarrow M_{2 \times 2}$ by $T(A) = A - A^T$. You can verify that this is a linear transformation.

(a) Find a basis for the kernel of T .

(b) What's the dimension of the image of T ?

(c) Find a basis for the image of T .

$$(a) A \in \ker T \Rightarrow T(A) = A - A^T = 0 \Rightarrow A = A^T.$$

$$\text{So } A \text{ is symmetric matrix.} \Rightarrow \ker T = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$$

$$(b) \text{ By Dimension thm, } \dim(\operatorname{Im} T) = 4 - 3 = 1.$$

$$(c). \quad T(A) = A - A^T = \begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 0 & b-c \\ c-b & 0 \end{bmatrix} = (b-c) \cdot \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

$$\Rightarrow \operatorname{Im} T = \operatorname{Span} \left\{ \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}.$$

Chapter 44: Isomorphisms and Composition

One-to-One and Onto Transformations

Let $T : V \rightarrow W$ be a linear transformation.

- T is said to be **onto** if $\text{im } T = W$ (i.e. $\dim \text{im } T = \dim W$).
- T is said to be **one-to-one** if $\mathbf{v}_1 \neq \mathbf{v}_2$ implies that $T(\mathbf{v}_1) \neq T(\mathbf{v}_2)$. An equivalent statement is that T is one-to-one if $T(\mathbf{v}_1) = T(\mathbf{v}_2)$ implies that $\mathbf{v}_1 = \mathbf{v}_2$.

Remark: Some books use the word “surjective” instead of onto, and “injective” instead of one-to-one.

Example: Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by

$$T \left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \right) = \begin{bmatrix} u_1 + u_3 \\ u_2 \end{bmatrix}.$$

Is T a one-to-one linear transformation?

$$\text{No. } T \left(\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right) = T \left(\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right)$$

Is T an onto linear transformation?

$$\text{Yes, } \forall \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in \mathbb{R}^2, \quad \exists \begin{bmatrix} u_1 \\ u_2 \\ 0 \end{bmatrix} \in \mathbb{R}^3, \text{ s.t.}$$

$$T \left(\begin{bmatrix} u_1 \\ u_2 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}.$$

Theorem: Let $T: V \rightarrow W$ be a linear transformation. Then T is one-to-one if and only if $\ker T = \{\mathbf{0}_V\}$.

Proof:

" $\Rightarrow$ " Assume T is 1-1, for $\forall v \in \ker T$,

$$\Rightarrow T(v) = T(\mathbf{0}) \Rightarrow v = \mathbf{0} \Rightarrow \ker T = \{\mathbf{0}\}.$$

" $\Leftarrow$ " Assume $\ker T = \{\mathbf{0}\}$, for $\forall v_1, v_2 \in V$,

$$\text{if } T(v_1) = T(v_2) \Rightarrow T(v_1 - v_2) = \mathbf{0}_W$$

$$\Rightarrow v_1 - v_2 \in \ker T \Rightarrow v_1 - v_2 = \mathbf{0}$$

$$\Rightarrow v_1 = v_2$$

so T is 1-1.

Corollary: Let $T: V \rightarrow W$ be a linear transformation. Then T is one-to-one if and only if $\dim \ker T = \text{nullity}(T) = 0$.

Example: Is $T: M_{23} \rightarrow M_{32}$ by $T(A) = A^T$ a one-to-one linear transformation?

$$\text{Yes. If } T(A) = A^T = B^T = T(B)$$

$$\Rightarrow A = B.$$

Example: Is $S: M_{22} \rightarrow \mathbb{R}$ by $S(A) = \text{tr}(A)$ is a one-to-one linear transformation?

$$\text{No. Let } A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$S(A) = S(B) = 0, \text{ but } A \neq B.$$

Recall: Any linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be written as a matrix transformation T_A where A is a $m \times n$ matrix.

Theorem: Let A be an $m \times n$ matrix, and let $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the linear transformation induced by A , that is $T_A(\mathbf{x}) = A\mathbf{x}$ for all columns $\mathbf{x}$ in $\mathbb{R}^n$.

1. T_A is onto if and only if $\text{rank } A = \underline{m}$ (here need $m \leq n$)
2. T_A is one-to-one if and only if $\text{rank } A = \underline{n}$ (here need $n \leq m$)

Example: Suppose $T : M_{22} \rightarrow \mathbb{R}^3$ is a linear transformation.

Can T be one-to-one?

No. $\dim(\text{Im } T) + \underbrace{\dim(\text{Ker } T)}_{=0} = 4$

$\Rightarrow \dim(\text{Im } T) = 4$, but $\dim(\text{Im } T) \leq \underbrace{\dim(W)}_3$

Can T be onto?

Yes. $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$

Example: Suppose $T : P_2 \rightarrow \mathbb{R}^4$ is a linear transformation.

Can T be one-to-one?

Yes. $T(ax^2 + bx + c) = \begin{bmatrix} a \\ b \\ c \\ 0 \end{bmatrix}.$

Can T be onto?

No. $\underbrace{\dim(\text{Im } T)}_{\substack{= \\ \dim(\mathbb{R}^4) \\ = \\ 4}} + \dim(\text{Ker } T) = 3 \Rightarrow \dim(\text{Ker } T) = -1.$

Contradiction!

Theorem: If $T : V \rightarrow W$ is a linear transformation and $\dim V = \dim W$, then

(a) If T is one-to-one, then T is onto.

(b) If T is onto, then T is one-to-one.

Proof: (Hint: Use the Dimension Theorem)

$$(a) \quad 1-1 \Rightarrow \dim(\ker T) = 0 \Rightarrow \dim(\operatorname{Im} T) = \dim(W)$$

$$(b) \quad \text{onto} \Rightarrow \dim(\operatorname{Im} T) = \dim(V) \Rightarrow \dim(\ker T) = 0.$$

Isomorphisms

What you might be noticing is that vector spaces may be represented differently, but act the same.

Example: How do we add two polynomials $ax + b$ and $cx + d$ in P_1 ?

$$(a+c) \cdot x + (b+d)$$

How do we add two column vectors $\begin{bmatrix} a \\ b \end{bmatrix}$ and $\begin{bmatrix} c \\ d \end{bmatrix}$ in $\mathbb{R}^2$?

$$\begin{bmatrix} a+c \\ b+d \end{bmatrix}$$

Definition: A linear transformation $T : V \rightarrow W$ is called an **isomorphism** if it is both onto and one-to-one. The vector spaces V and W are said to be **isomorphic** if there exists an isomorphism $T : V \rightarrow W$, and we write $V \simeq W$ when this is the case.

Remark: Other books may use a different symbol to say “isomorphic”: for example, $V \cong W$.

Example: $T : P_1 \rightarrow \mathbb{R}^2$ by $T(ax + b) = \begin{bmatrix} a \\ b \end{bmatrix}$ is an isomorphism.

Since T is 1-1 and onto.

Warning: This is not the only isomorphism possible! Try $L : P_1 \rightarrow \mathbb{R}^2$ by $L(ax + b) = \begin{bmatrix} b \\ a \end{bmatrix}$

Example: Show that M_{22} and $\mathbb{R}^4$ are isomorphic.

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}.$$

T is 1-1 and onto.

Example: Are M_{22} and P_2 isomorphic?

No. $\dim(M_{22}) \neq \dim(P_2)$

Example: Let V be a vector space with $\dim V = n$. The coordinate representation $C_B : V \rightarrow \mathbb{R}^n$ relative to an ordered basis B in V is an isomorphism between V and $\mathbb{R}^n$.

Theorem: If V and W are finite dimensional spaces, the following conditions are equivalent for a linear transformation $T : V \rightarrow W$.

1. T is an isomorphism.
2. If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is any basis of V , then $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$ is a basis of W .
3. There is at least one basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ of V such that $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$ is a basis of W .

Proof: ((1) implies (2)):

(1) $\Rightarrow$ (2) : If T is isom., then T is 1-1 and onto,
 $\Rightarrow W = \text{span}\{T\mathbf{v}_1, \dots, T\mathbf{v}_n\} \Rightarrow T\mathbf{v}_1 \dots T\mathbf{v}_n$ are L.I.
 $\Rightarrow \{T\mathbf{v}_1 \dots T\mathbf{v}_n\}$ is a basis.

Theorem: If V and W are finite dimensional vector spaces, then V and W are isomorphic if and only if $\dim V = \dim W$.

Proof: (if time)

Let $\{\mathbf{v}_1 \dots \mathbf{v}_n\}$ be a basis for V
 $\{\mathbf{w}_1 \dots \mathbf{w}_n\}$ be a basis for W .

Then $T : V \rightarrow W$ is an isom.

$$\mathbf{v}_i \rightarrow \mathbf{w}_i$$

Warning: If V, W have the same dimension, then there must exist an isomorphism between them. That does NOT mean all linear transformations between V and W are isomorphisms.

Example: Consider the linear transformation $T : P_1 \rightarrow \mathbb{R}$ by $T(ax + b) = \begin{bmatrix} a+b \\ 0 \end{bmatrix}$. Is this an isomorphism?

$$\text{No. } \underbrace{\dim(P_1)}_{2} \neq \underbrace{\dim(\mathbb{R})}_{1}.$$

Composition and Inverses

Definition: If $T : V \rightarrow W$ and $S : W \rightarrow U$ are linear transformations, the **composite** $ST : V \rightarrow U$ of T and S is defined as $ST(\mathbf{v}) = S(T(\mathbf{v}))$ for all $\mathbf{v}$ in V . The operation of forming the function ST is called **composition**.

Example: If $T : P_1 \rightarrow P_1$ by $T(ax + b) = (b + a)x - a$ and $S : P_1 \rightarrow P_1$ by $S(ax + b) = bx$, find ST and TS .

$$\begin{aligned} ST(ax+b) &= S((b+a)x - a) \\ &= -ax \\ TS(ax+b) &= T(bx) = bx - b. \end{aligned}$$

Recall: If V is a vector space, the function $I_V : V \rightarrow V$ by $I_V(\mathbf{v}) = \mathbf{v}$ is called the **identity operator**.

Fact: I_V is a linear transformation (In fact, it's an isomorphism).

Theorem: Let $T : V \rightarrow W$, $S : W \rightarrow U$, and $R : U \rightarrow Y$ be linear transformations.

1. The composite ST is again a linear transformation.

2. $TI_V = \underline{T}$ and $I_W T = \underline{T}$.

3. $(RS)T = \underline{R(ST)}$

Proof: (of (1)):

$$\forall v_1, v_2 \in V, \quad ST(v_1 + v_2) = S(Tv_1 + Tv_2) \\ = STv_1 + STv_2$$

$$ST(cv_1) = S(T(cv_1)) = S(cTv_1) = c \cdot S(Tv_1) \\ = c \cdot ST(v_1)$$

Theorem: Let V and W be finite dimensional vector spaces. The following conditions are equivalent for a linear transformation $T : V \rightarrow W$.

1. T is an isomorphism.

2. There exists a linear transformation $S : W \rightarrow V$ such that $ST = I_V$ and $TS = I_W$.

In this case S is also an isomorphism and is uniquely determined by T : If $\mathbf{w}$ in W is $\mathbf{w} = T(\mathbf{v})$, then $S(\mathbf{w}) = \mathbf{v}$. S is called the **inverse** of T and is denoted by T^{-1} .

Example: Consider $S : P_1 \rightarrow P_1$ by $S(ax + b) = bx$. Is S an isomorphism? If so, find S^{-1} .

No. $S(x+1) = S(2x+1) = x$, but
 $x+1 \neq 2x+1$

Example: Consider $T : P_1 \rightarrow P_1$ by $T(ax + b) = (b+a)x - a$. Is T an isomorphism? If so, find T^{-1} .

Yes. $T^{-1}T(ax+b) = T^{-1}((b+a)x - a)$
 $= ax+b$

$$\Rightarrow T^{-1}(ax+b) = -bx + (a+b)$$